

## MINA SALALAH, Oman

WMO: 413120

Lat: 16.934N

Lon: 54.008E

Elev: 82

StdP: 14.65

Time Zone: 4.00 (E04)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 65.9       | 67.7 | 27.9                          | 21.9 | 75.0 | 32.0 | 26.6 | 79.7 | 24.2                  | 73.1 | 20.9 | 74.3 | 6.0                   | 290  | 0.289 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 6.6                    | 93.0            | 65.0 | 90.8 | 66.7 | 89.1 | 69.9 | 82.8                | 86.1 | 81.9 | 85.8 | 81.2 | 85.3 | 7.1                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.8                            | 166.2 | 85.3 | 80.8 | 160.4 | 84.9 | 80.0 | 156.0 | 84.5 | 47.0          | 86.3 | 45.9 | 86.0 | 45.1 | 85.5 | 88.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| 17.9              | 14.8 | 12.3 | 61.3                       | 95.9 | 2.3                | 3.3 | 59.7                                               | 98.3 | 58.4       | 100.3 | 57.1       | 102.1 | 55.5       | 104.6 |    |    |
|                   |      |      | 47.3                       | 82.5 | 3.1                | 1.4 | 45.1                                               | 83.5 | 43.3       | 84.3  | 41.5       | 85.1  | 39.3       | 86.1  |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 79.2   | 74.7  | 75.5  | 78.7  | 82.3  | 85.2  | 83.6  | 77.8  | 75.9  | 78.9  | 80.0  | 80.2  | 77.7 |
|                                                                               | DBStd       | 4.10   | 2.76  | 3.22  | 3.44  | 2.51  | 2.09  | 2.63  | 2.43  | 1.72  | 2.90  | 2.62  | 2.56  | 2.78 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD50       | 10667  | 766   | 715   | 889   | 970   | 1091  | 1009  | 862   | 804   | 866   | 931   | 905   | 859  |
|                                                                               | CDD65       | 5192   | 301   | 295   | 424   | 520   | 626   | 559   | 397   | 339   | 416   | 466   | 455   | 394  |
|                                                                               | CDH74       | 48395  | 1619  | 1949  | 3994  | 6098  | 8139  | 6701  | 2655  | 1348  | 3320  | 4691  | 4719  | 3162 |
|                                                                               | CDH80       | 13791  | 140   | 267   | 969   | 2136  | 3701  | 2498  | 243   | 18    | 530   | 1284  | 1408  | 597  |
| Wind                                                                          | WSAvg       | 5.2    | 6.7   | 5.8   | 4.9   | 4.8   | 5.3   | 5.4   | 4.1   | 3.5   | 4.7   | 4.3   | 5.3   | 7.3  |
| Precipitation                                                                 | PrecAvg     | 4.6    | 0.2   | 0.1   | 0.3   | 0.3   | 0.5   | 0.2   | 0.9   | 0.7   | 0.3   | 0.2   | 0.4   | 0.4  |
|                                                                               | PrecMax     | 10.9   | 0.5   | 0.4   | 1.7   | 3.0   | 6.3   | 2.2   | 2.2   | 1.6   | 2.7   | 2.0   | 3.8   | 1.8  |
|                                                                               | PrecMin     | 1.3    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 2.3    | 0.1   | 0.1   | 0.5   | 0.5   | 1.3   | 0.4   | 0.5   | 0.4   | 0.5   | 0.4   | 0.7   | 0.4  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 85.5  | 88.1  | 93.6  | 94.5  | 95.5  | 94.9  | 86.1  | 82.1  | 93.4  | 91.8  | 91.8  | 88.7 |
|                                                                               |             | MCWB   | 60.6  | 61.5  | 63.2  | 64.2  | 66.3  | 78.7  | 80.2  | 74.4  | 65.5  | 64.2  | 63.2  | 62.1 |
|                                                                               | 2%          | DB     | 83.5  | 85.4  | 89.9  | 91.1  | 92.3  | 90.7  | 84.0  | 80.1  | 88.2  | 88.9  | 89.3  | 86.4 |
|                                                                               |             | MCWB   | 60.6  | 60.7  | 62.7  | 64.1  | 67.9  | 81.9  | 79.4  | 75.0  | 63.6  | 66.4  | 63.1  | 61.6 |
|                                                                               | 5%          | DB     | 81.6  | 83.4  | 87.0  | 88.6  | 89.9  | 88.5  | 82.4  | 79.1  | 84.3  | 87.1  | 87.3  | 84.7 |
|                                                                               |             | MCWB   | 62.5  | 61.1  | 63.0  | 67.9  | 74.1  | 81.5  | 78.6  | 75.0  | 69.4  | 68.7  | 64.9  | 62.5 |
|                                                                               | 10%         | DB     | 80.0  | 81.2  | 84.5  | 86.8  | 88.5  | 87.0  | 81.1  | 78.2  | 82.6  | 85.4  | 85.5  | 83.2 |
|                                                                               |             | MCWB   | 65.2  | 65.4  | 67.9  | 73.8  | 76.7  | 80.9  | 77.7  | 74.8  | 73.0  | 71.4  | 68.1  | 64.7 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 72.8  | 75.2  | 78.9  | 83.3  | 85.2  | 83.5  | 80.6  | 78.9  | 80.0  | 79.6  | 78.5  | 75.0 |
|                                                                               |             | MCDB   | 78.7  | 79.5  | 81.1  | 85.1  | 86.3  | 87.0  | 83.4  | 80.7  | 81.5  | 84.6  | 84.0  | 81.5 |
|                                                                               | 2%          | WB     | 71.1  | 73.5  | 77.5  | 80.8  | 83.4  | 82.7  | 79.6  | 77.4  | 78.4  | 78.5  | 76.8  | 73.5 |
|                                                                               |             | MCDB   | 78.1  | 79.0  | 81.7  | 85.0  | 86.5  | 86.0  | 82.2  | 78.9  | 81.1  | 83.9  | 83.0  | 80.7 |
|                                                                               | 5%          | WB     | 70.2  | 72.3  | 76.0  | 79.3  | 81.9  | 81.9  | 78.7  | 76.7  | 77.6  | 77.6  | 75.6  | 72.5 |
|                                                                               |             | MCDB   | 77.7  | 78.5  | 81.5  | 84.5  | 86.8  | 85.4  | 81.1  | 78.0  | 80.3  | 83.0  | 82.6  | 80.1 |
|                                                                               | 10%         | WB     | 69.3  | 71.1  | 74.7  | 78.5  | 81.1  | 81.4  | 77.8  | 76.2  | 77.0  | 76.8  | 74.6  | 71.5 |
|                                                                               |             | MCDB   | 77.2  | 77.9  | 80.9  | 84.1  | 86.3  | 85.0  | 79.9  | 77.5  | 79.8  | 82.1  | 82.1  | 79.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.5  | 10.6  | 10.1  | 8.6   | 6.6   | 5.1   | 3.6   | 3.5   | 5.2   | 9.5   | 11.4  | 10.8 |
|                                                                               |             | MCDBR  | 11.3  | 12.7  | 12.3  | 11.7  | 8.6   | 7.6   | 5.0   | 4.6   | 7.1   | 11.7  | 13.1  | 12.1 |
|                                                                               |             | MCWBR  | 6.3   | 7.4   | 6.7   | 5.6   | 3.4   | 2.8   | 2.1   | 2.1   | 2.8   | 6.1   | 7.1   | 5.7  |
|                                                                               | 5% WB       | MCDBR  | 9.6   | 8.8   | 8.3   | 7.0   | 5.7   | 4.6   | 4.3   | 4.0   | 5.4   | 8.3   | 9.9   | 9.4  |
|                                                                               |             | MCWBR  | 5.9   | 6.0   | 5.9   | 4.6   | 3.0   | 2.4   | 2.0   | 2.1   | 3.0   | 6.4   | 7.4   | 6.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.414  | 0.436 | 0.485 | 0.513 | 0.602 | 0.778 | 0.824 | 0.756 | 0.607 | 0.452 | 0.432 | 0.415 |      |
|                                                                               | taud        | 2.220  | 2.144 | 2.008 | 1.974 | 1.785 | 1.491 | 1.451 | 1.559 | 1.796 | 2.162 | 2.202 | 2.218 |      |
|                                                                               | Ebn at Noon | 275    | 275   | 265   | 257   | 232   | 192   | 184   | 199   | 230   | 267   | 267   | 271   |      |
|                                                                               | Edh at Noon | 43     | 49    | 57    | 60    | 71    | 94    | 98    | 89    | 70    | 47    | 43    | 42    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1732   | 1989  | 2210  | 2339  | 2339  | 2082  | 1772  | 1794  | 1970  | 2037  | 1781  | 1662  |      |
|                                                                               | RadStd      | 46     | 52    | 79    | 63    | 82    | 106   | 98    | 62    | 88    | 52    | 64    | 54    |      |

## Historical Trends

|                        | DBAvg      | Heating    |            | Cooling    |            |            | Degree-Days |            |            |            |
|------------------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|------------|
|                        |            | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD50       | HDD65      | CDD50      | CDD65      |
| Station Only           | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A  | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A |
| Regional (4 neighbors) | +0.27      | N/A        | -2.74      | N/A        | N/A        | -0.54      | N/A         | N/A        | N/A        | N/A        |

Nomenclature: See separate page